Prathamesh Nikam

AI Engineer / Research Engineer

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SUMMARY

M.Tech IT graduate combining AI research with application engineering. Built vision-language retrieval experiments, LangGraph multi-agent systems, and a published distributed deep-learning framework; experienced in comparative evaluation, model integration, and cloud deployment.

TECHNICAL SKILLS

AI & Models: LLMs, LangChain, LangGraph, Multi-Agent Systems, RAG, OpenCLIP, PyTorch, FAISS
Research Methods: Comparative Benchmarks, Ablation Studies, OOD Detection, Ensemble Learning, FedAvg, Quantization
Engineering: Python, SQL, NumPy, Pandas, Ray, FastAPI, REST APIs, GCP Cloud Run, Docker, Git

RESEARCH & PROJECTS

EvidenceMem | Visual Memory and Continual Classification [Link]
Python, PyTorch, OpenCLIP, FAISS, NumPy, MiniBatchKMeans
- Built a compact medoid memory over frozen CLIP embeddings with visual-text fusion, retrieved image evidence, and new-class insertion without encoder updates.
- Reported 89.58% mean CIFAR-10 accuracy across three seeds with 200 stored prototypes, 4.33 percentage points above zero-shot CLIP.
- Evaluated CIFAR-100 classification, OOD detection, memory budgets, and component ablations; documented the accuracy gap to a trained linear probe.

Neural Consensus Engine | Multi-Agent AI System [Link]
Python, LangGraph, Gemini, FastAPI, GCP Cloud Run
- Orchestrated Creative, Logical, and Ethical LLM agents in parallel, with a synthesis stage that compares outputs and produces a consensus response.
- Built FastAPI services, configurable prompts and temperatures, and workflow visualization; deployed on GCP Cloud Run.

HQDE | Hierarchical Quantized Distributed Ensemble [Link]
Python, PyTorch, Ray, ResNet-18, FedAvg
M.Tech dissertation, University of Hyderabad | June 2026 | Supervisor: Prof. P.S.V.S. Sai Prasad
- Developed and published a distributed deep-learning framework with stateful Ray workers, independent ensemble training, and synchronized FedAvg-style updates.
- Benchmarked five image datasets: independent HQDE exceeded centralized accuracy on four, with a largest observed gain of 2.62 percentage points.
- Implemented blockwise delta quantization with error feedback, reducing model-update communication by approximately 3.05x versus matched FP32 FedAvg.

EXPERIENCE

University of Hyderabad | Teaching Assistant - Web Development (IT) Lab | Jan - Jun 2026
- Guided students on project design and full-stack application development; recommended GenAI tools for specific project needs, suggested improvements, and supported project evaluation.

University of Hyderabad | Teaching Assistant - Java OOP Lab | Jul - Dec 2025
- Designed lab questions, evaluated student work, and guided students through Java object-oriented programming concepts and debugging.

EDUCATION

University of Hyderabad | M.Tech, Information Technology | 2024 - 2026 | CGPA: 8.68
Modern Education Society College of Engineering | B.E., Computer Engineering | 2019 - 2023
Honors in AI/ML | B.E. CGPA: 8.85

ACHIEVEMENTS

Data Science Conference Hackathon: Winner (2026) | AI4Bharat AWS Hackathon: Finalist | SupeRR Selection Hackathon: Top 36 of 500+ participants (2025) | Finalist in 10+ hackathons held across India.